EXISTENCE OF THE DERIVED TENSOR FUNCTOR AND ITS BASE-CHANGE ISOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Existence and computation of right derived functors on triangulated categories. Derived Categories, Section \ref{derived-section-derived-functors}, Situation \ref{derived-situation-derived-functor}, Definition \ref{derived-definition-right-derived-functor-defined}, and Lemma \ref{derived-lemma-find-existence-computes}, cited at proof lines 14679-14715. Provides the framework in which the derived tensor functor is constructed and shown to be everywhere defined.
- The total tensor product functor on complexes and its homotopy and exactness properties. Lemmas \ref{lemma-derived-tor-homotopy} and \ref{lemma-derived-tor-exact}, cited at proof line 14686. Identify the functor F whose derived functor is being constructed and give its basic properties.
- Existence of K-flat resolutions and their behaviour under base change and quasi-isomorphism. Lemmas \ref{lemma-K-flat-resolution}, \ref{lemma-derived-tor-quasi-isomorphism-other-side}, and \ref{lemma-base-change-K-flat}, cited at proof lines 14692, 14701, and 14721. Verify the hypotheses of the existence lemma and compute the derived functor on the complexes that appear in the base-change isomorphism.

Source reference. Original lemma 06Y6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8875–8875. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05S7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7436–7436. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7867–7867. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7868–7868. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05S8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7437–7437. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05S9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7438–7438. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06XN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8857–8857. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06Y4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8873–8873. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7870–7870. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05SW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7460–7460. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06Y1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8870–8870. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *The construction above is independent of choices and defines an exact functor of triangulated categories* $- \otimes^{\mathbf{L}}_R N^\bullet : D(R) \to D(A)$. *There is a functorial isomorphism*

$$E^\bullet \otimes^{\mathbf{L}}_R N^\bullet = (E^\bullet \otimes^{\mathbf{L}}_R A) \otimes^{\mathbf{L}}_A N^\bullet$$

for $E^\bullet$ *in* $D(R)$.

Proof. To prove the existence of the derived functor $- \otimes^{\mathbf{L}}_R N^\bullet$ we use the general theory developed in Derived Categories, Section 05S7. Set $\mathcal{D} = K(R)$ and $\mathcal{D}' = D(A)$. Let us write $F : \mathcal{D} \to \mathcal{D}'$ the exact functor of triangulated categories defined

by the rule $F(M^\bullet) = \mathrm{Tot}(M^\bullet \otimes_R N^\bullet)$. To prove the stated properties of F use Lemmas 064I and 064J. We let S be the set of quasi-isomorphisms in $\mathcal{D} = K(R)$. This gives a situation as in Derived Categories, Situation 05S8 so that Derived Categories, Definition 05S9 applies. We claim that LF is everywhere defined. This follows from Derived Categories, Lemma 06XN with $\mathcal{P} \subset \mathrm{Ob}(\mathcal{D})$ the collection of K-flat complexes: (1) follows from Lemma 06Y4 and (2) follows from Lemma 064L. Thus we obtain a derived functor

$$LF : D(R) = S^{-1}\mathcal{D} \longrightarrow \mathcal{D}' = D(A)$$

see Derived Categories, Equation (05SW). Finally, Derived Categories, Lemma 06XN guarantees that $LF(K^\bullet) = F(K^\bullet) = \mathrm{Tot}(K^\bullet \otimes_R N^\bullet)$ when $K^\bullet$ is K-flat, i.e., LF is indeed computed in the way described above. Moreover, by Lemma 06Y1 the complex $K^\bullet \otimes_R A$ is a K-flat complex of A-modules. Hence

$$(K^\bullet \otimes^{\mathbf{L}}_R A) \otimes^{\mathbf{L}}_A N^\bullet = \mathrm{Tot}((K^\bullet \otimes_R A) \otimes_A N^\bullet) = \mathrm{Tot}(K^\bullet \otimes_A N^\bullet) = K^\bullet \otimes^{\mathbf{L}}_A N^\bullet$$

which proves the final statement of the lemma. □